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Renewable energy reduces domestic depression but increases depression for neighboring countries: evidence of spatial effects from 181 countries worldwide

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Abstract

Background Depression poses a significant global public health challenge, affecting millions of people worldwide. The utilization of renewable energy holds potential for improving mental health by reducing air pollution and promoting green spaces.

Purpose This study aims to investigate the impact of renewable energy use on depression, with a focus on its spatial effects and the mediating roles of air pollution reduction and green space expansion.

Methods Data from 181 countries were analyzed using a two-way fixed effects model and the Spatial Durbin Model (SDM). Depression-related metrics, including Disability-Adjusted Life Years (DALYs), Age-Standardized Disability-Adjusted Life Years Rate (ASDR), prevalence, and Age-Standardized Prevalence Rate (ASPR), were evaluated.

Results Renewable energy use significantly reduces DALYs, ASDR, prevalence, and ASPR within a country, but it also significantly increases the risk of depression in neighboring countries. The impact of renewable energy on depression varies by gender, age, and SDI level, being more pronounced for males and the 50–74 age group. The effect is significant in high and low SDI countries but not in middle SDI countries, indicating a "middle-income trap."

Conclusion Renewable energy can improve mental health by reducing air pollution and promoting green spaces. However, policymakers need to consider spatial effects and tailor policies accordingly to maximize health benefits.

Keywords Renewable energy, Depression, Environmental pollution, Green space, Spatial effects, DALYs, ASDR, ASPR

Introduction

In recent years, the global prevalence of depression has significantly increased, becoming a major public health issue [51, 78]. According to the World Health Organization (WHO), over 300 million people worldwide are affected by depression, making it one of the leading causes of disability and disease burden [43]. The high incidence of depression not only affects individuals' quality of life and lifespan but also has substantial socio-economic impacts [39, 57]. It is estimated that depression-related productivity losses and healthcare costs impose a burden of over $1 trillion annually on the

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global economy [15, 45]. Against this global trend, there is a growing body of research focusing on the impact of environmental factors on mental health.

Role of renewable energy in mental health improvement

The United Nations Sustainable Development Goals (SDGs) provide a framework to address these challenges. SDG 3 aims to ensure healthy lives and promote well-being for all ages, with the prevention and treatment of depression being a crucial component of this goal [71]. Additionally, SDG 7 emphasizes ensuring access to affordable, reliable, sustainable, and modern energy for all [47]. Promoting the use of renewable energy not only aids in environmental protection and reducing carbon emissions but also contributes to creating a healthier living environment, thereby supporting improvements in mental health [73, 75]. Specifically, renewable energy usage, as a key means of enhancing environmental quality, has shown potential positive effects in alleviating depression [74]. Studies indicate that renewable energy use helps reduce pollution and improve living conditions, which may indirectly enhance mental health by increasing green spaces and promoting outdoor activities [28]. Therefore, exploring the relationship between renewable energy use and depression is of significant importance. Addressing the global challenge of depression through sustainable energy usage not only meets environmental protection requirements but also has profound public health implications, reflecting the integrated and synergistic nature of the SDGs [52, 73, 75].

The relationship between renewable energy use and depression is significant. Firstly, using renewable energy helps reduce air pollution, which in turn lowers depression rates [55]. Studies have shown that prolonged exposure to high levels of air pollution increases the risk of depression and other mental health issues [80]. By utilizing solar, wind, and hydro power, harmful pollutants in the air are significantly reduced, thereby decreasing the incidence of depression among residents [25]. Secondly, the use of renewable energy is often accompanied by an increase in green spaces and improvements in the ecological environment, which have positive effects on mental health [69]. Psychological studies suggest that exposure to natural environments and green spaces can reduce stress levels, enhance mood, and strengthen psychological resilience. Green spaces offer more opportunities for outdoor activities, encouraging physical exercise and social interaction, which have been proven to prevent and alleviate depression [67].

According to the Restorative Environments Theory (RET), reducing air pollution aligns with the principle of "reduced interference." Air pollution is a significant environmental stressor, and by using renewable energy to reduce pollution, the risk of mental health issues, including depression, is lowered [14]. Additionally, the increase in green spaces and improvements in the ecological environment have restorative effects, not only reducing stress levels and enhancing mood but also strengthening psychological resilience [37]. RET emphasizes that natural environments can restore attention, alleviate mental fatigue, and improve overall psychological health [8]. The outdoor activities and social interactions facilitated by green spaces provide opportunities for "vitality restoration," which helps increase positive emotions and reduce feelings of loneliness and depression [49].

National differences and mechanisms of impact

Existing studies have shown that the use of renewable energy positively impacts health conditions such as depression. However, these effects may vary significantly across different geographic regions. For instance, Nordic countries like Denmark and Norway, which extensively use wind and hydro power, not only enjoy higher air quality but also have relatively better mental health levels among their residents [38]. In contrast, countries like India and China, which primarily rely on coal, still face severe air pollution problems despite recent increases in renewable energy use, resulting in higher rates of depression and other mental health issues [29]. Additionally, California in the United States, California, a leading region in solar energy use, has experienced improvements in air quality, which have been shown to positively affect residents' mental health [30]. By analyzing a broader range of regions, we can verify and understand the spatial effects of renewable energy use, providing scientific evidence for global energy policies and public health strategies.

The use of renewable energy can significantly alter various pollution indicators, which are closely related to human physiological indicators, thereby effectively reducing the risk of depression [62]. Firstly, using renewable energy sources such as solar, wind, and hydro power reduces reliance on fossil fuels, significantly lowering concentrations of particulate matter (e.g., PM2.5 and PM10), nitrogen oxides (NOx), sulfur dioxide (SO2), and volatile organic compounds (VOCs) in the air [23]. The reduction of these pollutants directly decreases oxidative stress and chronic inflammation levels in the body, reducing the release of pro-inflammatory cytokines (e.g., IL-6 and TNF-α), thereby lowering neuroinflammatory responses associated with depression [3, 19]. Moreover, reducing air pollutants helps maintain the balance of neurotransmitters, especially serotonin and dopamine, which play crucial roles in regulating mood and preventing depression [9]. Additionally, reducing coal-fired power generation decreases environmental emissions of heavy metals such as mercury, lead, and cadmium, which are toxic to the nervous and endocrine systems [11].

Reducing exposure to these metals helps maintain neural health [54]. Furthermore, the development of renewable energy is often accompanied by an increase in green spaces and improvements in the ecological environment [59]. Exposure to natural environments and green spaces not only increases the synthesis of vitamin D in the body and reduces cortisol levels, a stress hormone, but also promotes outdoor activities and social interactions, enhancing psychological resilience and emotional state [50]. These ecological improvements help reduce stress and anxiety, further alleviating symptoms of depression [56].

Research gaps and study objectives

While existing research has preliminarily revealed the potential impact of renewable energy on mental health, studies on the specific effects and mechanisms of renewable energy use on depressive symptoms remain relatively limited [44]. Current research often focuses on the direct impact of environmental pollution on depression, lacking in-depth exploration of the indirect effects and spatial effects of renewable energy use [9]. This study aims to fill this gap by analyzing the spatial effects of renewable energy use and the mediating role of environmental pollution, exploring its unique contribution to depressive symptoms. Our research will provide a new perspective, highlighting the importance of environmental improvement in enhancing mental health [35, 61].

Based on the aforementioned background, this study proposes the following hypotheses: increasing the proportion of renewable energy use significantly improves depressive symptoms. Global and Local Moran's I are employed to assess spatial autocorrelation and clustering distribution among countries. Additionally, the Spatial Durbin Model (SDM) is utilized to measure the spatial effects of renewable energy on depression. Renewable energy use indirectly improves depressive symptoms by reducing environmental pollution. By testing these hypotheses, we aim to reveal the effects and mechanisms of renewable energy use on improving depressive symptoms, providing scientific evidence for the formulation and implementation of public health policies.

Methods

Empirical model

To assess the impact of renewable energy use on depression and the heterogeneity of these effects across different groups, we employ a two-way fixed effects model. The model is specified as follows:

$$y_{it} = \alpha + \beta Renewable\ energy_{it} + \delta X_{it} + \mu_i + \gamma_t + \varepsilon_{it} \quad (1)$$

where, y_{it} represents a range of proxy variables regarding depression. $Renewable\ energy_{it}$ represents the proportion of renewable energy use in the total final energy consumption. The coefficient β reflects the influence degree that renewable energy use on depression. X_{it} represents a series of country-level control variables. μ_i and γ_t are fixed effects for country and year, respectively, and ε_{it} is a random disturbance term.

Given that depressive symptoms have become a significant global health issue and that economic, social, cultural, and geographic factors may introduce spatial correlation among countries, we further analyze whether the impact of renewable energy use on depression exhibits spatial effects. First, before conducting the spatial effect analysis, it is necessary to perform a spatial autocorrelation test on the dependent variable to determine the appropriateness of using spatial econometric models [58, 76]. Typically, global and local Moran's I statistics are used for this purpose, which can be expressed as follows:

$$Global\ Moran's\ I = \frac{n\sum_{i=1}^{n}\sum_{j=1}^{n} W_{ij}\left(y_i - \bar{y}\right)\left(y_j - \bar{y}\right)}{\sum_{i=1}^{n}\sum_{j=1}^{n} W_{ij} \times \sum_{i=1}^{n}\left(y_i - \bar{y}\right)^2} \quad (2)$$

$$Local\ Moran's\ I = \frac{n\left(y_i - \bar{y}\right)\sum_{j=1}^{n} W_{ij}\left(y_j - \bar{y}\right)}{\sum_{i=1}^{n}\left(y_i - \bar{y}\right)^2} \quad (3)$$

Among the variables, n represents the number of countries in this study, y_i and y_j is the depression variable for each country, and $\bar{y}$ denotes the average of the depression variables across all countries. W_{ij} is the spatial weight matrix, which we constructed using the straight-line distances between the capitals of the respective countries. The Global Moran's I in Eq. (2) reflects the degree of spatial autocorrelation across the entire region: $I \in [-1,1]$ indicates positive spatial correlation; $I < 0$ indicates negative spatial correlation; and $I = 0$ suggests no spatial correlation. The Local Moran's I in Eq. (3) captures the spatial heterogeneity between countries, representing the patterns of spatial clustering that exist in different regions.

We employed spatial econometric models to analyze whether the use of renewable energy has a spatial effect on depression, including the Spatial Durbin Model (SDM), Spatial Autoregressive Model (SAR), and Spatial Error Model (SEM) [7, 17]. We determined the most suitable model through likelihood ratio (LR) test and Wald test, and the Hausman test was used to decide between fixed and random effects. The expressions for the aforementioned spatial models are as follows:

$$SDM: y_{it} = \alpha + \rho\sum W_{ij}y_{jt} + \beta\sum X_{it} + \delta\sum W_{ij}X_{it} + \mu_i + \gamma_t + \varepsilon_{it} \quad (4)$$

Table 1 Descriptive statistics for each variable

Variables	Definition	Unit	Obs	Mean	Std. Dev.	Min	Max
DALYs	Disability-adjusted life years	100,000 year	5430	2.129	7.622	0.0004	87.295
ASDR	Age-standardized Disability-adjusted life years rate	year per 100,000 people	5430	662.751	193.181	254.273	1486.692
Prevalence	Number of people suffering from depression	100,000 people	5430	12.699	46.774	0.003	520.053
ASPR	Age-standardized Prevalence rate	people per 100,000 people	5430	3882.548	999.628	1605.277	8146.385
Renewable energy	Ratio of renewable energy in total energy consumption	-	5430	0.118	0.136	0	0.816
CO_2 emissions	Total CO_2 emissions	Metric ton	5430	154.413	685.309	0	10,721.04
NO_2 exposure	NO_2 content in the air	ppb	5430	5.422	3.804	0.232	22.515
Ambient PM exposure	Particulate matter content in the air	ug/m^3	5430	26.967	15.989	5.127	107.145
Green space	The ratio of forest area to land area	-	5430	0.345	0.243	0.0001	0.986
PGDP	Gross domestic product per capita	constant 10,000 LCU	5430	192.608	1164.533	0.023	17,681.779
RIAV	Ratio of industrial added value to GDP	-	5430	0.268	0.122	0.024	0.860
TP	Total population	10,000 people	5430	3550.818	13,199.480	0.918	141,000
HW	Health workers	people per 10,000 people	5430	120.118	129.955	6.138	696.085
SDI	Socio-demographic index	-	5430	0.558	0.191	0.073	0.929

To avoid heteroscedasticity, we take logarithmic treatment of DALYs, ASDR, Prevalence, ASPR, CO_2 emissions, NO_2 exposure, ambient PM exposure, PGDP, TP and HW in subsequent empirical analysis

$$SAR: y_{it} = \alpha + \rho \sum W_{ij} y_{jt} + \beta \sum X_{it} + \mu_i + \gamma_t + \varepsilon_{it} \quad (5)$$

$$SEM: y_{it} = \alpha + \beta \sum X_{it} + \mu_i + \gamma_t + \varepsilon_{it} \quad \varepsilon_{it} = \lambda \sum W_{ij} \varepsilon_{it} + \nu_{it} \quad (6)$$

Variables

Dependent variable

Depression. We use data from the GBD database covering depression statistics from 1990 to 2019 for various countries. The proxies include Disability-adjusted life years (DALYs), Prevalence, Age-standardized Disability-adjusted life years rate (ASDR), and Age-standardized Prevalence rate (ASPR) [13, 32]. DALYs aims to measure the health impact of depression by counting all healthy years of life lost from onset to death. ASDR adjusts DALYs according to the standard age distribution of the population to eliminate the influence of the difference in the age structure of the population in different regions and different years. Prevalence measures the distribution of people with depression in a country. ASPR is to adjust the Prevalence indicator according to the standard age distribution of the population. Using these proxies allows for a comprehensive reflection of the national depression situation.

Independent variable

Renewable Energy. We measure this by the proportion of renewable energy in total final energy consumption [36]. According to the International Energy Agency (IEA), renewable energy includes Solar PV, Hydroelectricity, Wind, and Bioenergy.

Mediator variable

Environmental Pollution and Green Space. We use CO_2 emissions, NO_2 exposure, and ambient PM exposure as proxies for environmental pollution [18, 60, 68]. Using the ratio of forest area to land area as proxy for green space [16].

Control variables

Based on existing literature, we control for economic level, industrial structure, population size, healthcare capacity, and the socio-demographic index (SDI) [12, 79]. Economic level is measured by GDP per capita (PGDP), industrial structure by the ratio of industrial added value to GDP (RIAV), population size by total population (TP), and healthcare capacity by health workers (HW).

Data and sources

Data on depression, NO_2 exposure, ambient PM exposure, HW, and SDI are sourced from the Global Health Data Exchange (GHDx) website (http://ghdx.healthdata.org/gbd-results-tool). Renewable energy data is obtained from the IEA's SDG7 Database (https://www.iea.org/). CO_2 emissions data comes from the Global Carbon Atlas (https://globalcarbonatlas.org/). Data on PGDP, RIAV, TP and green space is from the World Bank database (https://databank.worldbank.org/). Descriptive statistics of the variables are presented in Table 1. Among them,

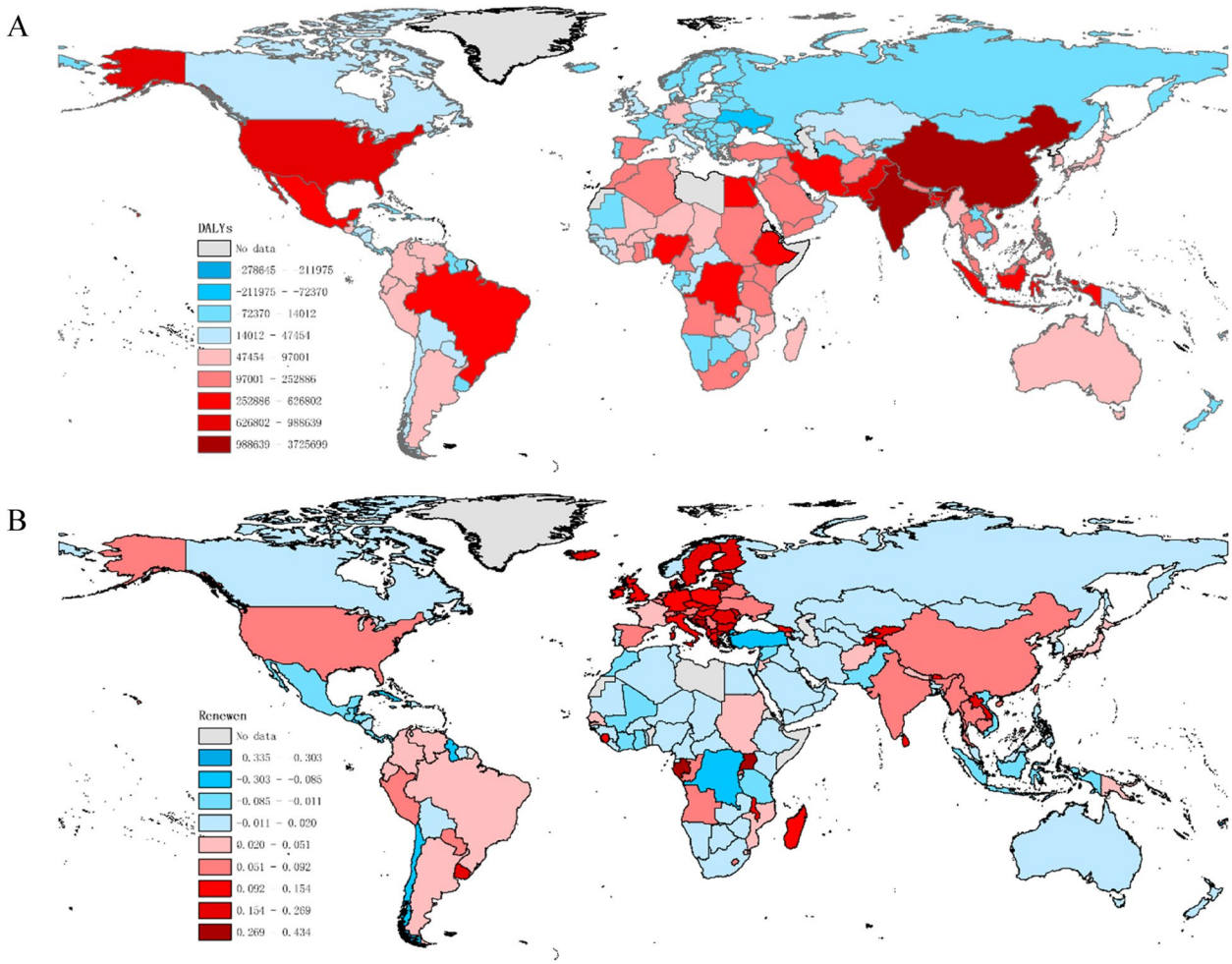

Fig. 1 The degree of change in core variables between 1990 and 2019 (**A**: DALYs; **B**: Renewable energy)

renewable energy, green space, RIAV and SDI are all in the form of ratio, and other indicators are original values. But it should be noted that in the specific model analysis, logarithmic transformations are performed on DALYs, ASDR, Prevalence, ASPR, CO_2 emissions, NO_2 exposure, ambient PM exposure, PGDP, TP and HW.

Figures 1 and S1 illustrate the changes in depression and renewable energy usage across countries from 1990 to 2019. The results show significant improvements in depression rates in Nordic countries like Denmark, Finland, and Latvia over the past 30 years, accompanied by an over 20% increase in the share of renewable energy. However, countries like China, India, and Ethiopia, due to their large and rapidly growing populations, have seen notable increases in DALYs and Prevalence, while ASDR and ASPR have shown significant decreases, alongside an increase in the share of renewable energy.

Results

Regression analysis of the effect of renewable energy on depression

Before conducting the regression analysis, we performed tests for multicollinearity and stationarity. The Variance Inflation Factor (VIF) values in Table S1 are all below 10, and the results in Table S2 are significant at the 5% level, indicating that the data do not suffer from multicollinearity issues and are stationary.

The results in Table 2 show that as the proportion of renewable energy usage increases, there is a significant improvement in the overall depression situation of the population. Specifically, for every 1% increase in the proportion of renewable energy usage, there is a significant reduction of 0.257% in DALYs, 0.088% in ASDR, 0.249% in Prevalence, and 0.071% in ASPR. This indicates that the use of renewable energy helps to reduce the overall level of depression in the population.

Table 2 The effects of renewable energy on depression

Variables	Ln(DALYs)	Ln(ASDR)	Ln(Prevalence)	Ln(ASPR)
Renewable energy	−0.257*** (0.017)	−0.088*** (0.013)	−0.249*** (0.016)	−0.071*** (0.011)
Ln(PGDP)	−0.049*** (0.004)	−0.021*** (0.003)	−0.049*** (0.004)	−0.019*** (0.003)
RIAV	0.015 (0.017)	0.009 (0.013)	0.014 (0.016)	0.009 (0.011)
Ln(TP)	0.897*** (0.008)	0.031*** (0.006)	0.880*** (0.008)	0.020*** (0.005)
Ln(HW)	0.070*** (0.008)	−0.010 (0.006)	0.078*** (0.007)	−0.007 (0.005)
SDI	0.612*** (0.050)	0.118** (0.038)	0.602*** (0.046)	0.114*** (0.032)
Country fixed effect	√	√	√	√
Year fixed effect	√	√	√	√
Obs	5430	5430	5430	5430
R^2	0.919	0.159	0.933	0.161

, * represent significance levels of 0.05, 0.01 and 0.001, respectively

Spatial effects of renewable energy on depression

First, the Global Moran's I calculated for DALYs, ASDR, Prevalence, and ASPR based on Eq. (2) are all significantly negative, indicating spatial autocorrelation in depression across countries rather than a random distribution (see Table S3).

Second, the Local Moran's I calculated based on Eq. (3) shows that the spatial clustering patterns of the depression proxies have remained relatively stable over the 30 years. More countries are located in the second quadrant (low–high) and the fourth quadrant (high-low), indicating an uneven and negatively correlated spatial distribution of depression across countries (see Fig. 2).

Building on this, we further analyzed whether the use of renewable energy has spatial effects on depression through a spatial econometric model. The results in Table S4 indicate that the Hausman test supports the use of a fixed-effect model, and both the LR and Wald tests suggest that the Spatial Durbin Model (SDM) is more appropriate. Therefore, we used the fixed-effect SDM for our analysis.

The direct and indirect effects calculated using the partial differentiation method show that renewable energy usage significantly reduces depression within the country [40]. Specifically, a 1% increase in the proportion of renewable energy usage leads to reductions of 0.182% in DALYs, 0.056% in ASDR, 0.175% in Prevalence, and 0.045% in ASPR. However, an increase in renewable energy usage within a country also significantly exacerbates depression in neighboring countries, with a 1% increase leading to increases of 1.524% in DALYs, 0.534% in ASDR, 1.724% in Prevalence, and 0.448% in ASPR (see Table 3).

Overall, the impact of renewable energy usage on depression exhibits spatial effects, reducing depression within the country while worsening it in neighboring countries.

The mediating role of environmental pollution and green space

We used a structural equation model (SEM) to analyze the mediating role of environmental pollution (CO_2 emissions, NO_2 exposure, and ambient PM exposure) and green space in the effect of renewable energy on depression. The results in Fig. 3 indicate that an increase in the proportion of renewable energy improves national depression levels by reducing environmental pollution and increasing green space.

Specifically, a 1% increase in the proportion of renewable energy usage significantly reduces CO_2 emissions by 1.086%, NO_2 exposure by 0.704%, ambient PM exposure by 0.451% and increases green space by 3.6%. Taking DALYs as an example, the mediating effect of CO_2 emissions is −0.033 (accounting for 12.89%), the mediating effect of NO_2 exposure is −0.023 (accounting for 8.98%), the mediating effect of ambient PM exposure is −0.016 (accounting for 6.25%), and the mediating effect of green space is −0.009 (accounting for 3.52%).

Heterogeneity analysis by gender, age, and SDI level

We further explored the differential impact of renewable energy on depression across different groups based on gender, age, and SDI levels. The results in Table 4 indicate that the use of renewable energy has a more significant effect on improving depression among males compared to females.

Regarding age groups, renewable energy usage shows a more pronounced improvement in depression for the 50–74 age group, while it does not exhibit a significant impact on the 0–14 age group.

For different SDI categories, the impact of renewable energy on national depression levels is not significant within the Middle SDI category, indicating the presence of a "middle-income trap".

Discussion

The core findings of this study indicate that increasing the proportion of renewable energy usage significantly improves the depressive symptoms of the national population, with notable regional variations in these effects. Specifically, renewable energy usage not only reduces air pollution and environmental stress but also indirectly improves residents' mental health by promoting green spaces. This study highlights the potential positive impact

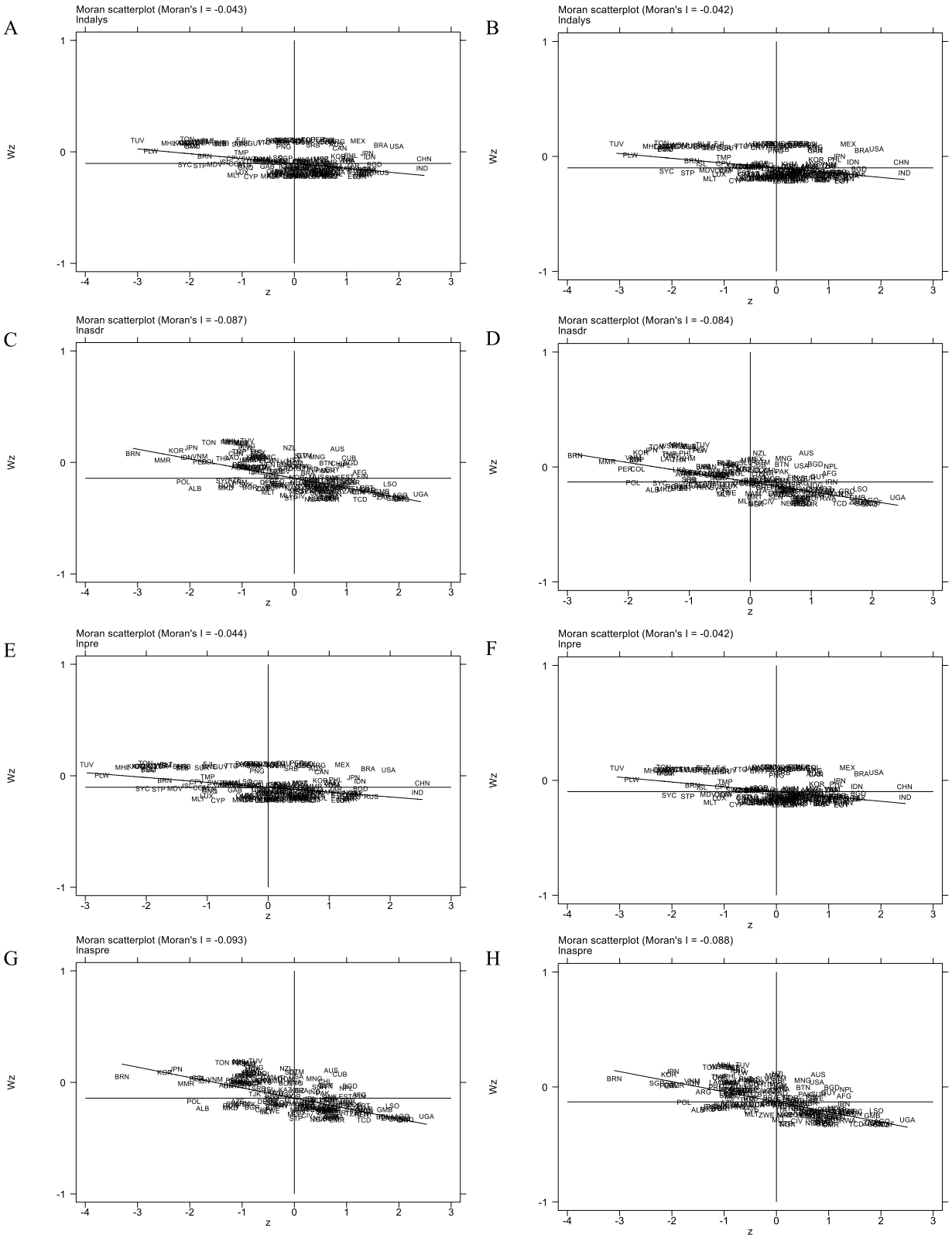

Fig. 2 Local Moran scatter plot of depression in various countries (**A**: DALYs in 1990; **B**: DALYs in 2019; **C**: ASDR in 1990; **D**: ASDR in 2019; **E**: Prevalence in 1990; **F**: Prevalence in 2019; **G**: ASPR in 1990; **H**: ASPR in 2019)

Table 3 Direct and indirect effects of renewable energy on depression

Variables	Ln(DALYs)		Ln(ASDR)		Ln(Prevalence)		Ln(ASPR)	
	Direct	Indirect	Direct	Indirect	Direct	Indirect	Direct	Indirect
Renewable energy	−0.182*** (0.017)	1.524*** (0.354)	−0.056*** (0.013)	0.534* (0.252)	−0.175*** (0.016)	1.724*** (0.397)	−0.045*** (0.011)	0.448* (0.215)
Ln(PGDP)	−0.044*** (0.004)	0.045 (0.049)	−0.020*** (0.003)	−0.018 (0.044)	−0.044*** (0.004)	0.055 (0.051)	−0.018*** (0.003)	−0.013 (0.037)
RIAV	0.013 (0.016)	−0.157 (0.332)	0.009 (0.012)	0.481 (0.322)	0.009 (0.015)	−0.464 (0.345)	0.009 (0.010)	0.303 (0.267)
Ln(TP)	0.896*** (0.009)	0.040 (0.150)	0.016* (0.007)	−0.328* (0.147)	0.882*** (0.008)	0.053 (0.157)	0.008 (0.006)	−0.271* (0.125)
Ln(HW)	0.059*** (0.007)	−0.018 (0.121)	−0.008 (0.005)	0.001 (0.107)	0.065*** (0.006)	−0.131 (0.127)	−0.006 (0.005)	−0.001 (0.092)
SDI	0.722*** (0.045)	0.246 (1.253)	0.144*** (0.035)	1.724 (1.166)	0.714*** (0.041)	0.806 (1.328)	0.138*** (0.030)	1.517 (0.999)
ρ	−1.033*** (0.205)		−0.777*** (0.209)		−0.785*** (0.206)		−0.752*** (0.210)	

*, *** represent significance levels of 0.05, 0.01 and 0.001, respectively

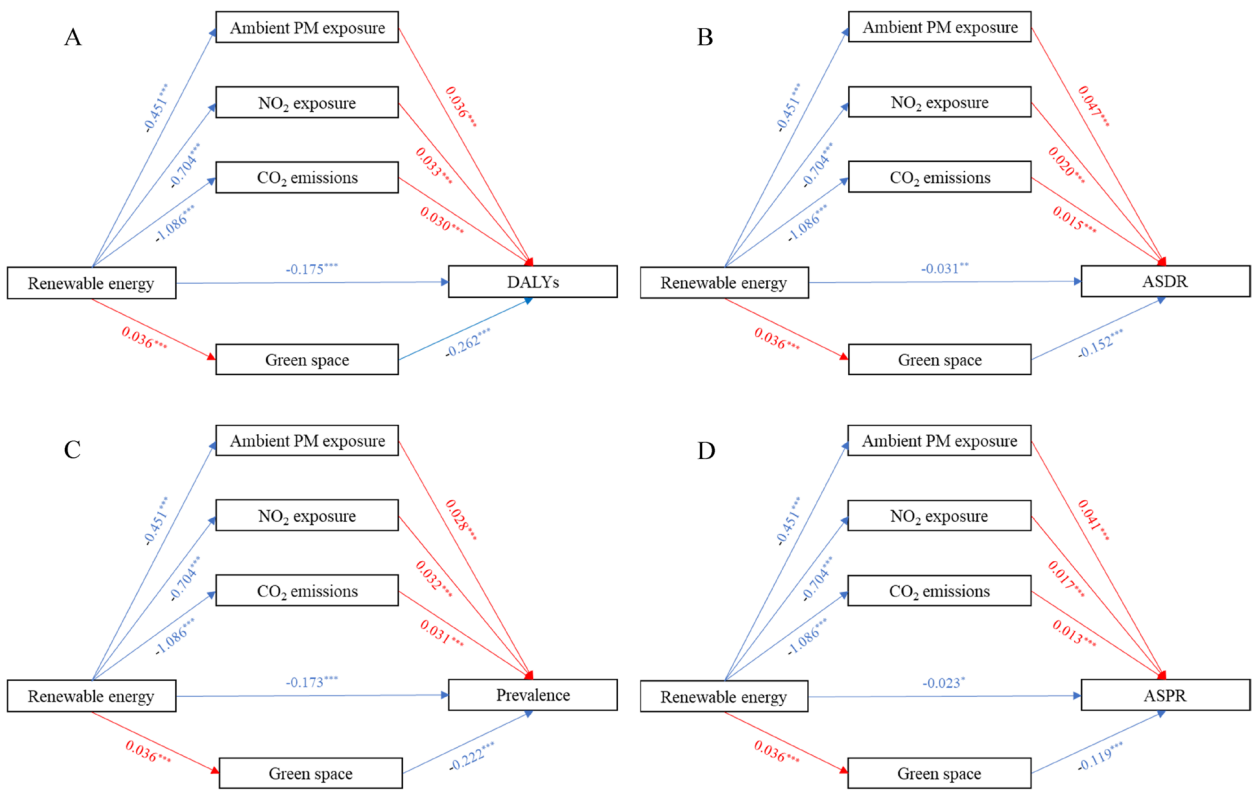

Fig. 3 Mediating role of CO_2 emissions, NO_2 exposure and ambient PM exposure (**A**: DALYs; **B**: ASDR; **C**: Prevalence; **D**: ASPR)

of renewable energy usage on mental health and underscores the importance of environmental improvement in public health policy [9].

First, our study finds that the use of renewable energy significantly improves overall depressive symptoms, consistent with existing research [41, 64]. Multiple studies have shown that reducing air pollution helps decrease the incidence of depression. For example, Singapore's extensive promotion of solar and clean energy technologies has successfully reduced urban air pollution levels, significantly improving residents' mental health [48, 70]. Additionally, China's recent efforts in developing renewable

Table 4 Summary of heterogeneous results

Category	Detail	Ln(DALYs)	Ln(ASDR)	Ln(Prevalence)	Ln(ASPR)
Gender	Male	−0.278*** (0.018)	−0.098*** (0.012)	−0.267*** (0.016)	−0.082*** (0.010)
	Female	−0.249*** (0.018)	−0.080*** (0.014)	−0.242*** (0.017)	−0.064*** (0.012)
Age	0–14 years	−0.016 (0.031)		−0.011 (0.030)	
	15–49 years	−0.233*** (0.021)		−0.215*** (0.020)	
	50–74 years	−0.574*** (0.028)		−0.564*** (0.028)	
	≥ 75 years	−0.196*** (0.038)		−0.200*** (0.037)	
SDI	High	−0.213** (0.065)	−0.278*** (0.063)	−0.170** (0.055)	−0.219*** (0.053)
	High-middle	−0.161*** (0.032)	−0.010 (0.026)	−0.161*** (0.029)	−0.004 (0.022)
	Middle	−0.016 (0.037)	0.026 (0.026)	−0.028 (0.033)	0.021 (0.022)
	Low-middle	−0.223*** (0.040)	−0.102*** (0.027)	−0.209*** (0.037)	−0.081*** (0.023)
	Low	−0.240*** (0.028)	−0.079*** (0.020)	−0.229*** (0.026)	−0.067*** (0.017)

Depression data for different gender as well as age groups are obtained from GBD. The classification of SDI is also based on the GBD official practice, which defines 0–0.455 as Low, 0.455–0.608 as Low-middle, 0.608–0.690 as Middle, 0.690–0.805 as High-middle, 0.805–1 as High

, * represent significance levels of 0.05, 0.01 and 0.001, respectively

energy have reduced coal usage, improved air quality, and led to a decline in depression rates in some urban areas [26, 42]. In summary, increasing the proportion of renewable energy usage significantly improves the national population's depressive symptoms by directly reducing air pollution and indirectly enhancing emotional well-being and psychological resilience through increased green spaces.

Second, our study confirms the spatial effects of renewable energy on depression levels across different countries. Renewable energy usage not only reduces depressive symptoms within a country but also has a significant impact on neighboring countries, leading to increased depression-related indicators in those countries. Renewable energy technologies have certain barriers and siphoning effects [27]. Extensive use of renewable energy in one country reduces air pollution and environmental stress, but neighboring countries may not benefit equally due to technological barriers, potentially exacerbating depressive symptoms due to technological lag [63]. Economic competition and energy dependency also play crucial roles. A country's large-scale shift to renewable energy may reduce demand for fossil fuels, increasing economic pressure on neighboring countries that rely on fossil fuel exports, leading to social instability and mental health issues [66]. Pollution transfer is another important factor. Reducing fossil fuel usage in one country may lead to the relocation of related industries to neighboring countries, increasing their environmental pollution and health risks [73, 75]. For instance, Germany's vigorous promotion of renewable energy and reduction of coal usage might negatively impact coal-exporting countries like Poland, affecting their economic and social stability [1, 20]. These complex factors jointly explain why renewable energy usage can improve mental health within a country but potentially have negative effects on neighboring countries.

Third, our structural equation model (SEM) analysis reveals that increasing renewable energy usage can indirectly improve national depression levels by reducing environmental pollution (such as CO_2 emissions, NO_2 exposure, and PM exposure and increasing green space). Existing environmental and biological research supports this mediation mechanism. Reducing CO_2 emissions decreases the accumulation of greenhouse gases, thereby reducing environmental stress associated with depression [33]. Studies have shown that high CO_2 concentrations are linked to increased psychological stress and anxiety symptoms [46]. NO_2 is a common air pollutant, and long-term exposure to high NO_2 levels can lead to respiratory diseases, inflammatory responses, and nervous system damage [31]. Biological studies indicate that NO_2 exposure

can increase oxidative stress and inflammation, disrupting normal nervous system function and increasing depression risk [10]. PM exposure is another significant environmental risk factor. Fine particulate matter (PM2.5) and coarse particulate matter (PM10) can penetrate deep into the lungs and bloodstream [21], causing chronic inflammation and oxidative stress [5]. Studies have shown that PM exposure is closely associated with the incidence and severity of depressive symptoms [77]. Furthermore, increasing green spaces provide recreational areas that encourage physical activity, reduce stress, and improve overall mental well-being [53]. Access to green spaces has been shown to lower the risk of depression and anxiety by providing a natural environment that promotes relaxation and social interaction [22]. Additionally, beyond the direct physiological benefits of reducing pollution and increasing green space, several studies have demonstrated that green spaces may provide a restorative effect on mental health through mechanisms such as stress reduction and improved mood regulation. For example, exposure to green environments has been shown to lower cortisol levels, a marker of stress, and increase heart rate variability, an indicator of better autonomic nervous system function. Furthermore, green spaces contribute to reduced neuroinflammation by lowering the levels of circulating pro-inflammatory cytokines, which have been implicated in the pathophysiology of depression.

The improvement in mental health through renewable energy is achieved by reducing environmental stress and promoting physical and mental well-being through the reduction of pollution and the increase of green spaces. The magnitude of these coefficients reflects the contribution of each mediation path to improving mental health, with the mediating effects of CO2 emissions and NO2 exposure being the most significant, highlighting their importance in alleviating mental health burdens.These interactions highlight the complex yet potent effects of environmental interventions on mental health, underscoring the need for integrated policies that promote both pollution reduction and increased access to green spaces as part of public health strategies.Additionally, socioeconomic disparities often impact access to these green spaces, with disadvantaged communities having less access to recreational areas. Promoting equitable access to green spaces, particularly in low-income urban areas, can help reduce mental health inequalities.Overall, reducing CO_2 emissions, NO_2 exposure, PM exposure and increasing green space can lower bodily inflammation and oxidative stress, thereby reducing neuroinflammatory responses associated with depression and improving mental health.

Finally, our study finds that the impact of renewable energy usage on depressive symptoms varies by gender, age group, and SDI (Socio-Demographic Index) levels. The improvement effect is more significant for males and the 50–74 age group, possibly because these groups play important roles in economic and social activities, facing higher work stress and environmental exposure [24]. Environmental improvements such as better air quality and optimized living conditions directly reduce environmental stress and health risks for these groups, significantly improving their mental health. In contrast, the impact on the 0–14 age group is not significant, possibly because children and adolescents in this age group mainly live and study indoors, with less exposure to external environmental pollution, and their mental health is more influenced by family, education, and social factors [6, 72]. Among different SDI categories, the impact of renewable energy usage on depression is significant in high SDI and low SDI countries but not in middle SDI countries, indicating a "middle-income trap." High SDI countries have well-developed infrastructure and environmental policies, effectively promoting and applying renewable energy, significantly improving environmental quality and living conditions, thus enhancing mental health [34]. Although low SDI countries have poorer economic conditions, severe environmental pollution issues mean any environmental improvement brings significant health benefits [65]. Middle SDI countries may be in a transitional phase of economic development and environmental governance, unable to effectively apply renewable energy like high SDI countries and facing environmental issues similar to low SDI countries, but the improvement is insufficient to bring significant mental health benefits [2].

Implications

Based on our findings, we propose several policy recommendations to promote renewable energy use and improve global mental health:

Promote renewable energy use

Increasing the proportion of renewable energy is an effective way to improve mental health. Reducing CO_2 emissions, NO_2 exposure, PM exposure and increasing green space can directly enhance air quality and indirectly improve residents' mental health by lowering environmental stress and inflammation. Policymakers should invest more in and promote renewable energy technologies, especially in areas with severe air pollution, prioritizing clean energy projects such as solar, wind, and hydropower [34].

Tailor policies for different demographics

Policies should be differentiated by gender and age groups. Our research shows that males and the 50–74

age group benefit more from renewable energy use, likely due to higher environmental stress in economic and social activities [4]. Therefore, policies should focus on these high-risk groups, improving work and community environments to further reduce depression risk. For the 0–14 age group, although the direct benefits are not significant, mental health can still be improved through education and family support [6].

Targeted policies based on SDI levels

Policies should be tailored to countries with different SDI levels. In high SDI countries, policies should continue to promote renewable energy use, enhance infrastructure, and improve environmental measures to consolidate and expand health benefits. In low SDI countries, addressing severe environmental pollution should be a priority. International aid and technical support can accelerate the adoption of clean energy and reduce the negative health impacts of air pollution [28]. For middle SDI countries, avoiding the "middle-income trap" requires coordinated energy and environmental policies. Improving energy efficiency and environmental protection measures can achieve dual goals of economic development and health improvement [20].

Enhance international cooperation

Policymakers should emphasize cross-national cooperation to coordinate regional energy and environmental policies, avoiding negative effects from renewable energy technology barriers and economic competition. Establishing regional cooperation mechanisms to share renewable energy technologies and experiences can reduce conflicts from pollution transfer and resource competition, promoting common development and health improvement in the region [64].

Overall, this study provides scientific evidence for formulating environmental and public health policies, emphasizing the importance of renewable energy use in improving mental health. By considering the needs of different groups and regions, targeted policies can more effectively enhance global mental health and advance sustainable development goals.

Limitations

This study highlights the significant impact and spatial effects of renewable energy use on depressive symptoms, but several limitations must be acknowledged. The data, while covering 181 countries, may lack sufficient temporal and regional coverage, particularly in developing countries with incomplete data. Mental health, being multifaceted, requires more comprehensive indicators beyond depressive symptoms for future analysis. Moreover, varying national policies on renewable energy use and depression prevention complicate cross-national comparisons. The study also assumes homogeneity in relationships across regions, which may overlook diverse socio-economic and environmental contexts. Additionally, potential omitted variable bias, such as unobserved cultural or institutional differences, may influence the findings. Future research should address these factors for a more nuanced understanding.

Conclusion

This study reveals that increasing the use of renewable energy significantly improves depressive symptoms across 181 countries, highlighting its positive impact on mental health. Renewable energy adoption reduces air pollution, promotes green spaces, all of which contribute to better mental health. The findings underscore the need for targeted policies to enhance renewable energy use, especially in high-pollution areas, and emphasize the importance of regional cooperation to maximize health benefits. This research provides a strong foundation for integrating renewable energy strategies into public health policies, aiming for both environmental sustainability and mental well-being.

Supplementary Information

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Supplementary Material 1

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Authors' contributions

Zhang Zheng was responsible for drafting the initial manuscript, Xu Huijie handled data processing and visualization, Li Ping Cui was responsible for organizing the raw data, responding to reviewer comments, and language polishing. Wang Yuanyuan was in charge of review and funding acquisition.

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Data availability

This study uses publicly available databases, and readers can download the data independently.

Declarations

Ethics approval and consent to participate

This study uses publicly available databases and does not require ethical approval.The data are in accordance with the Helsinki Declaration.

Competing interests

The authors declare no competing interests.

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